BUILDNEO

IN SEARCH OF INNOVATION

RSE

RIVER SEDIMENT EXPLORER

PRESENTED IN INMEX SMM-23

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IN SEARCH OF INNOVATION

BUILDNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

RIVER SEDIMENT EXPLORER

The project focuses on the development of a bathymetric Autonomous Surface Vehicle (ASV) with a catamaran-like hull design for use in Indian rivers. This vehicle utilizes SLAM (Simultaneous Localization and Mapping) based navigation algorithms to create predictive maps of siltation patterns. The primary goal is to enhance the effective navigational depth of Indian rivers.
Hydrostatic calculations for the hull forms have been carried out, and their models have been designed accordingly.The SLAM navigation algorithm has been tested and successfully implemented on a small-scale prototype.

OBJECTIVE

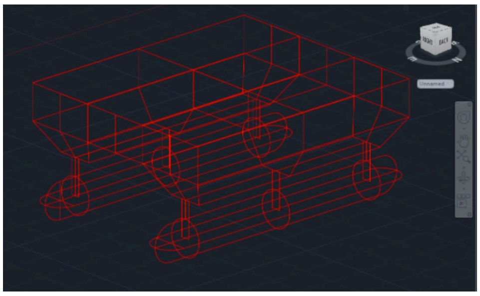
Development of a catamaran-hull ASV for bathymetric mapping in Indian rivers.
Utilizes SLAM-based navigation to create predictive siltation maps.
Aims to enhance navigational depth in rivers.

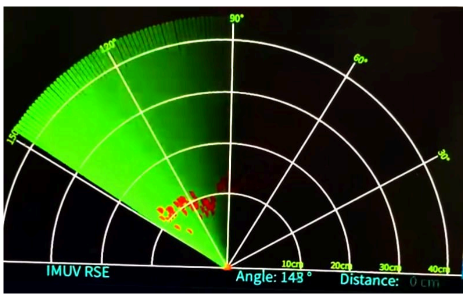
SPECIFICATIONS

- DOF: 6
- Main Hull: Custom Aluminum Milling
- Electronics Bay: Custom Aluminum Laser-cut
- Maximum dive depth: 24m
- LOA: 800mm
- Beam: 350mm

FEATURES

Catamaran Hull Design – Ensures stability, reduced resistance, and better maneuverability in river conditions.
SLAM-Based Navigation – Uses Simultaneous Localization and Mapping (SLAM) for autonomous movement and accurate riverbed mapping.
Bathymetric Data Collection – Equipped with depth sensors to measure and analyze siltation patterns.
Predictive Mapping – Generates real-time maps to assess sediment deposition and optimize navigational depth.
Autonomous Operation – Navigates independently without human intervention, improving efficiency and coverage.
Hydrostatic Optimization – Hull design is optimized based on hydrostatic calculations for enhanced buoyancy and performance.
Prototype Validation – SLAM algorithm successfully tested on a small-scale prototype, ensuring reliability before full-scale deployment.





FUTURE WORK

The next phase of this project involves the development of a SWATH (Small Waterplane Area Twin Hull) based Autonomous Surface Vehicle (ASV), as shown in the model above. This design offers superior stability compared to traditional catamaran hulls, making it ideal for precise bathymetric surveys in dynamic river environments.